

motion and Rates BOS

Part A: motion



NAME: _____

Q 1

2000

A particle is moving in a straight line, starting from the origin. At time t seconds the particle has a displacement of x metres from the origin and a velocity $v \text{ m s}^{-1}$. The displacement is given by $x = 2t - 3 \log_e(t + 1)$.

- (i) Find an expression for v .
- (ii) Find the initial velocity.
- (iii) Find when the particle comes to rest.
- (iv) Find the distance travelled by the particle in the first three seconds.

(a) (i) $x = 2t - 3 \log_e(t + 1)$

$$v = \frac{dx}{dt}$$

$$= 2 - \frac{3}{t+1}$$

(ii) Let $t = 0$, $v = 2 - 3 = -1$.
 $\therefore$ Initial velocity is -1 m s^{-1} .

(iii) Let $v = 0$, $2 - \frac{3}{t+1} = 0$
 $2t + 2 - 3 = 0$

$$t = \frac{1}{2}$$

$\therefore$ Particle is at rest when $t = \frac{1}{2} \text{ s}$.

At $t = 0$, $x = 0$.

At $t = \frac{1}{2}$, $x = 1 - 3 \ln 1.5$.

At $t = 3$, $x = 6 - 3 \ln 4$.

Distance travelled in the first 3 seconds

$$= |1 - 3 \ln 1.5 - 0| + |6 - 3 \ln 4 - (1 - 3 \ln 1.5)|$$

$$= 3 \ln 1.5 - 1 + 6 - 3 \ln 4 - 1 + 3 \ln 1.5$$

$$= 2.27 \text{ m.}$$

(iv) At $t = 0$, $x = 2(0) - 3 \ln 1 = 0$.

At $t = \frac{1}{2}$, $x = 2(0.5) - 3 \ln 1.5$
 $= 1 - 3 \ln 1.5$
 $= -0.2164$.

At $t = 3$, $x = 2 \times 3 - 3 \ln 4$
 $= 6 - 3 \ln 4$
 $= 1.8411$.

A	O	B
$t = \frac{1}{2}$	$t = 0$	$x = 1.8411$
$x = -0.2164$	$x = 0$	$t = 3$

Distance $OA = 0.2164 \text{ m}$.

Distance $OB = 1.8411 \text{ m}$.

Distance travelled $= 2OA + OB$

$$= 2 \times 0.2164 + 1.8411$$

$$= 2.2739 \dots$$

$$= 2.27 \text{ m (3 sig. figs).}$$

Q 2

2001

A particle moves in a straight line so that its displacement, in metres, is given by

$$x = \frac{t-2}{t+2} \quad \text{where } t \text{ is measured in seconds.}$$

- (i) What is the displacement when $t = 0$?
- (ii) Show that $x = 1 - \frac{4}{t+2}$.
 Hence find expressions for the velocity and the acceleration in terms of t .
- (iii) Is the particle ever at rest? Give reasons for your answer.
- (iv) What is the limiting velocity of the particle as t increases indefinitely?

(c) $x = \frac{t-2}{t+2}$

(i) When $t = 0$, $x = \frac{0-2}{0+2} = -1$.

(ii) $1 - \frac{4}{t+2} = \frac{t+2}{t+2} - \frac{4}{t+2}$
 $= \frac{t-2}{t+2}$

$$\therefore x = 1 - \frac{4}{t+2}$$

$$= 1 - 4(t+2)^{-1}$$

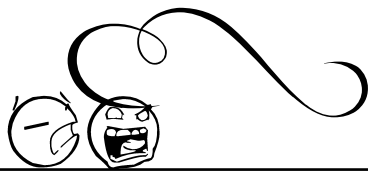
Velocity $= \frac{dx}{dt}$
 $= 0 - 4 \times (-1) \times (t+2)^{-2}$
 $= 4(t+2)^{-2}$
 $= \frac{4}{(t+2)^2}$ —①

Acceleration $= \frac{dv}{dt}$
 $= 4 \times (-2) \times (t+2)^{-3}$, from ①
 $= \frac{-8}{(t+2)^3}$

(iii) No. The particle is never at rest because
 $v = \frac{4}{(t+2)^2}$ is always greater than zero.

(iv) $\lim_{t \rightarrow \infty} \frac{4}{(t+2)^2} = 0$,
 $\therefore$ the limiting velocity is 0 m s^{-1} .





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Part A: motion



Q 3

2002

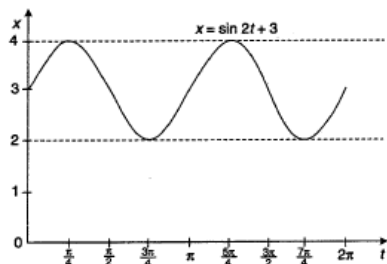
A particle moves in a straight line. At time t seconds, its distance x metres from a fixed point O on the line is given by

$$x = \sin 2t + 3.$$

- Sketch the graph of x as a function of t for $0 \leq t \leq 2\pi$.
- Using your graph, or otherwise, find the times when the particle is at rest, and the position of the particle at those times.
- Describe the motion completely.

(b) (i) Range: $-1 + 3 \leq x \leq 1 + 3$
 $\therefore 2 \leq x \leq 4$

Period: $\frac{2\pi}{2} = \pi$.



(ii) The particle is at rest when

$$\frac{dx}{dt} = 0, \text{ i.e. the gradient of the tangent is } 0.$$

Time Position

$$t = \frac{\pi}{4} \text{ s } \quad x = 4 \text{ m}$$

$$t = \frac{3\pi}{4} \text{ s } \quad x = 2 \text{ m}$$

$$t = \frac{5\pi}{4} \text{ s } \quad x = 4 \text{ m}$$

$$t = \frac{7\pi}{4} \text{ s } \quad x = 2 \text{ m}$$

- The particle oscillates between $x = 4$ and $x = 2$ with the centre of oscillation at $x = 3$. The particle starts at $x = 3$ and moves to the right to $x = 4$, where at $t = \frac{\pi}{4}$ s, the particle stops, changes direction and moves to the left, until at $x = 2$ and $t = \frac{3\pi}{4}$ s, it again stops, changes direction and moves to the right again, returning to the centre of oscillation at $x = 3$ ($t = \pi$ s). This motion is then repeated from $t = \pi$ s to $t = 2\pi$ s.

Q 4

2003

The velocity of a particle is given by

$$v = 2 - 4 \cos t \text{ for } 0 \leq t \leq 2\pi,$$

where v is measured in metres per second and t is measured in seconds.

- At what times during this period is the particle at rest?
- What is the maximum velocity of the particle during this period?
- Sketch the graph of v as a function of t for $0 \leq t \leq 2\pi$.
- Calculate the total distance travelled by the particle between $t = 0$ and $t = \pi$.

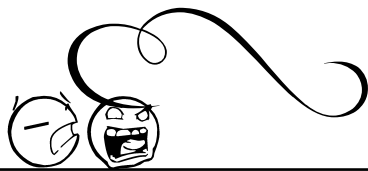
$v = 2 - 4 \cos t \rightarrow t = \pi$
 $v = 2 - 4 \cos 180^\circ$
 $= 2 + 4 = 6 \text{ m/s}$

(b) (i) $v = 2 - 4 \cos t$
 $0 = 2 - 4 \cos t$
 $4 \cos t = 2$
 $\cos t = \frac{1}{2}$ ✓
 $t = 60^\circ, 300^\circ$
 $= \frac{\pi}{3}, \frac{5\pi}{3}$ ✓

(ii) $4 \sin t = 0$
 $\sin t = 0$
 $t = 0, 180^\circ, 360^\circ$
 $= 0, \pi, 2\pi$ ✓

(iv) $A = \left| \int_0^{\frac{\pi}{3}} (2 - 4 \cos t) dt \right| + \left| \int_{\frac{\pi}{3}}^{\pi} (2 - 4 \cos t) dt \right|$
 $= 4\sqrt{3} + 2\pi$ ✓
 Need to show all working to obtain this result.

Quicker to look at the graph of $v = 2 - 4 \cos t$ to find the max. + p.



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Part A: motion



Q 5

2004

A particle moves along a straight line so that its displacement, x metres, from a fixed point O is given by $x = 1 + 3 \cos 2t$, where t is measured in seconds.

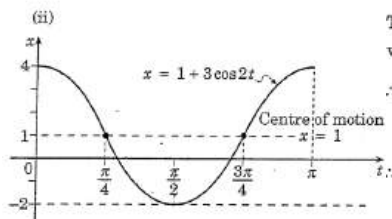
- What is the initial displacement of the particle?
- Sketch the graph of x as a function of t for $0 \leq t \leq \pi$.
- Hence, or otherwise, find when AND where the particle first comes to rest after $t = 0$.
- Find a time when the particle reaches its maximum speed. What is this speed?

(b) $x = 1 + 3 \cos 2t$

(i) Initial displacement when $t = 0$.

$$\begin{aligned} x &= 1 + 3 \cos 2(0) \\ &= 1 + 3 \cos 0 \\ &= 1 + 3 \times 1 \\ &= 4. \end{aligned}$$

$\therefore$ Initial displacement is 4 metres to the right of the point O .



(iii) The particle comes to rest when $v = 0$.

$$\begin{aligned} v &= \frac{dx}{dt} \\ &= -6 \sin 2t \\ 0 &= -6 \sin 2t \\ 0 &= \sin 2t \\ 2t &= 0, \pi, 2\pi, \dots \\ t &= 0, \frac{\pi}{2}, \pi, \dots \end{aligned}$$

The next time $v = 0$ after $t = 0$ will be when $t = \frac{\pi}{2}$ seconds.

$$\begin{aligned} \therefore x &= 1 + 3 \cos \left(2 \times \frac{\pi}{2} \right) \\ &= 1 + 3 \cos \pi \\ &= -2. \end{aligned}$$

$\therefore$ The particle first comes to rest after $t = 0$ when $t = \frac{\pi}{2}$ and $x = -2$,

(ie. when the particle is 2 metres to the left of point O).

(iv) The particle reaches maximum speed when $a = 0$, ie. maximum speed is reached when $\frac{dv}{dt} = 0$.

$$\begin{aligned} v &= -6 \sin 2t \\ \frac{dv}{dt} &= -12 \cos 2t. \end{aligned}$$

When $\frac{dv}{dt} = 0$, $0 = -12 \cos 2t$

$$\begin{aligned} \cos 2t &= 0 \\ 2t &= \frac{\pi}{2}, \frac{3\pi}{2}, \dots \\ t &= \frac{\pi}{4}, \frac{3\pi}{4}, \dots \end{aligned}$$

Taking $t = \frac{\pi}{4}$, $v = -6 \sin 2 \left(\frac{\pi}{4} \right) = -6 \text{ m/s}$

$\therefore$ Maximum speed = $|-6 \text{ m/s}| = 6 \text{ m/s}$.

$\therefore$ When $t = \frac{\pi}{4}$, the particle reaches its maximum speed of 6 m/s.

Q 6

2005

A particle is initially at rest at the origin. Its acceleration as a function of time, t , is given by

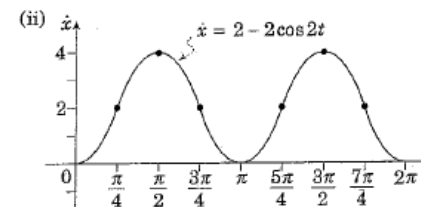
$$\ddot{x} = 4 \sin 2t.$$

- Show that the velocity of the particle is given by $\dot{x} = 2 - 2 \cos 2t$.
- Sketch the graph of the velocity for $0 \leq t \leq 2\pi$ AND determine the time at which the particle first comes to rest after $t = 0$.
- Find the distance travelled by the particle between $t = 0$ and the time at which the particle first comes to rest after $t = 0$.

(i) When $t = 0$, $\dot{x} = 0$ and $x = 0$:

$$\begin{aligned} \ddot{x} &= 4 \sin 2t \\ \dot{x} &= \int \ddot{x} \, dt \\ &= \int 4 \sin 2t \, dt \\ \dot{x} &= -2 \cos 2t + C. \end{aligned}$$

$$\begin{aligned} \text{When } t = 0, \dot{x} = 0: \quad 0 &= -2 \cos 2(0) + C \\ \therefore C &= 2 \\ \dot{x} &= 2 - 2 \cos 2t. \end{aligned}$$



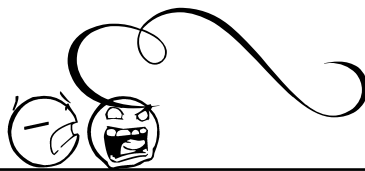
Period = $\frac{2\pi}{2} = \pi$. Amplitude = 2.

Particle first comes to rest (after $t = 0$) at $t = \pi$.

(iii) Distance travelled:

$$\begin{aligned} x &= \int_0^\pi (2 - 2 \cos 2t) \, dt \\ &= [2t - \sin 2t]_0^\pi \\ &= (2\pi - 0) - (0 - 0) \\ &= 2\pi. \end{aligned}$$

$\therefore$ Distance travelled is 2π units.



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Part A: motion



Q 7

2006

A particle is moving in a straight line. Its displacement, x metres, from the origin, O , at time t seconds, where $t \geq 0$, is given by $x = 1 - \frac{7}{t+4}$.

- Find the initial displacement of the particle.
- Find the velocity of the particle as it passes through the origin.
- Show that the acceleration of the particle is always negative.
- Sketch the graph of the displacement of the particle as a function of time.

(i) initial: $t = 0 \therefore x = 1 - \frac{7}{0+4}$
 $x = -\frac{3}{4}$

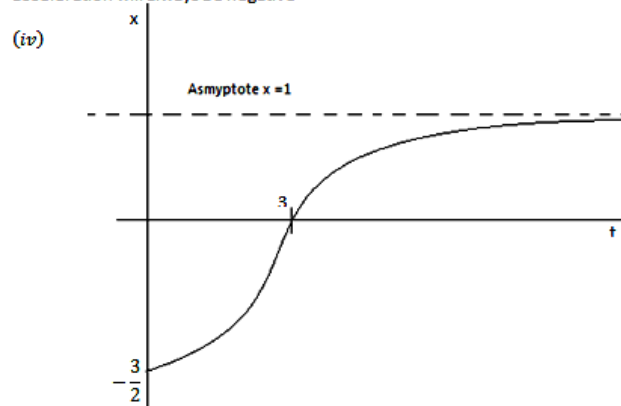
(ii) when $x = 0 \therefore 0 = 1 - \frac{7}{t+4}$
 $t+4 = 7$
 $\therefore t = 3$

(iii) $x = 1 - 7(t+4)^{-1}$

$$\dot{x} = \frac{7}{(t+4)^2}$$

$$\ddot{x} = -\frac{14}{(t+4)^3}$$

Since $t > 0$ denominator will always be positive, and since numerator is a constant negative the acceleration will always be negative



Q 8

2007

A particle is moving on the x -axis and is initially at the origin. Its velocity, v metres per second, at time t seconds is given by

$$v = \frac{2t}{16+t^2}$$

- What is the initial velocity of the particle?
- Find an expression for the acceleration of the particle.
- Find the time when the acceleration of the particle is zero.
- Find the position of the particle when $t = 4$.

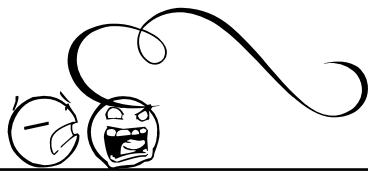
(b)(i) $v = \frac{2t}{16+t^2}$
 when $t=0 \therefore v=0$

(ii) $\frac{dv}{dt} = \frac{2(16+t^2)-4t^2}{(16+t^2)^2}$
 $a = \frac{32-2t^2}{(16+t^2)^2}$

(iii) $a = \frac{32-2t^2}{(16+t^2)^2}$
 $0 = \frac{32-2t^2}{(16+t^2)^2}$
 $0 = 32-2t^2$
 $t^2 = 16$
 $t = 4 (t > 0)$

(iv) $x = \int \frac{2t}{16+t^2} dt$
 $= \log_e(16+t^2) + c$
 when $t=0, x=0$
 $\therefore 0 = \log_e(16) + c$
 $c = -\log_e(16)$
 $x = \log_e(16+t^2) - \log_e(16)$
 when $t=4$,
 $x = \log_e(16+4^2) - \log_e(16)$
 $= \log_e\left(\frac{32}{16}\right) = \log_e(2) \text{ m}$





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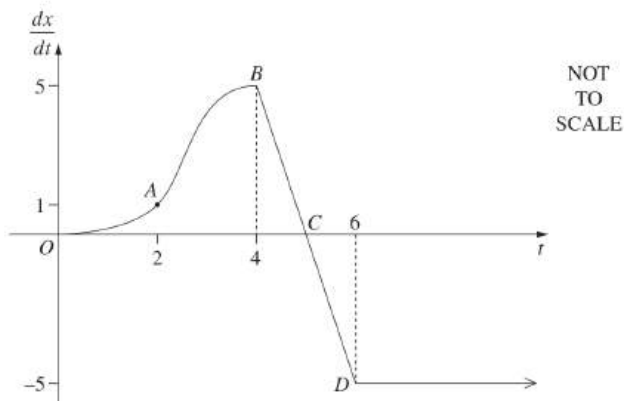
Part A: motion



Q 9

2007

An object is moving on the x -axis. The graph shows the velocity, $\frac{dx}{dt}$, of the object, as a function of time, t . The coordinates of the points shown on the graph are $A(2, 1)$, $B(4, 5)$, $C(5, 0)$ and $D(6, -5)$. The velocity is constant for $t \geq 6$.



- Using Simpson's rule, estimate the distance travelled between $t = 0$ and $t = 4$. 2
- The object is initially at the origin. During which time(s) is the displacement of the object decreasing? 1
- Estimate the time at which the object returns to the origin. Justify your answer. 2
- Sketch the displacement, x , as a function of time. 2

$$(a)(i) d = \frac{h}{3} [f(\text{first}) + 4f(\text{even}) + 2f(\text{odd}) + f(\text{last})]$$

$$= \frac{2}{3} [0 + 4(1) + 5]$$

$$= 6$$

(ii) displacement is decreasing when velocity is negative
 $\therefore t > 5$

(iii) returns to the origin when the area above the t axis on the velocity curve is equal to the area below the t axis.

$$t = 4 \text{ to } 5 \therefore x = \frac{5}{2}$$

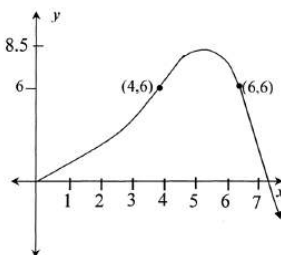
$$t = 5 \text{ to } t = 6 \therefore x = -\frac{5}{2}. \text{ These cancel out.}$$

$$t = 0 \text{ to } t = 4 \therefore x = 6$$

The particle needs to travel 6 units in the negative direction after $t = 6$ seconds

$$\therefore t = \frac{d}{s} = \frac{6}{5} = 1.2 \text{ seconds}$$

$\therefore$ it returns to the origin at 7.2 seconds



Q10

2008

The graph shows the velocity of a particle, v metres per second, as a function of time, t seconds.

- What is the initial velocity of the particle?
- When is the velocity of the particle equal to zero?
- When is the acceleration of the particle equal to zero?
- By using Simpson's Rule with five function values, estimate the distance travelled by the particle between $t = 0$ and $t = 8$.

Q6bi Read from graph, $v = 20 \text{ ms}^{-1}$.

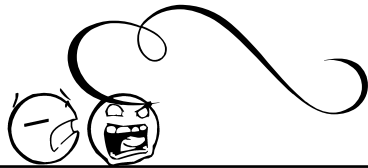
Q6bii Read from graph, $t = 10 \text{ s}$.

Q6biii Read from graph, $t = 6 \text{ s}$.

$$Q6biv \quad A = \frac{1}{3}(20 + 4(37) + 50) + \frac{1}{3}(50 + 4(61) + 70)$$

$$+ \frac{1}{3}(70 + 4(77) + 80) + \frac{1}{3}(80 + 4(75) + 60) \approx 493 \text{ m}$$





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Part A: motion



NAME: _____

Q11

2009

The acceleration of a particle is given by

$$\ddot{x} = 8e^{-2t} + 3e^{-t},$$

where x is displacement in metres and t is time in seconds.

Initially its velocity is -6 m s^{-1} and its displacement is 5 m .

- (i) Show that the displacement of the particle is given by $x = 2e^{-2t} + 3e^{-t} + t$.
- (ii) Find the time when the particle comes to rest.
- (iii) Find the displacement when the particle comes to rest.

Q7ai $\ddot{x} = 8e^{-2t} + 3e^{-t}$, $x = 5$ and $\dot{x} = -6$ when $t = 0$.

$\therefore \dot{x} = -4e^{-2t} - 3e^{-t} + 1$ and $x = 2e^{-2t} + 3e^{-t} + t$.

Q7aii The particle comes to rest: $\dot{x} = -4e^{-2t} - 3e^{-t} + 1 = 0$
 $(-4e^{-t} + 1)(e^{-t} + 1) = 0$.

Since $e^{-t} + 1 > 0$, $\therefore -4e^{-t} + 1 = 0$, $e^{-t} = \frac{1}{4}$, $e^t = 4$, $t = \ln 4$.

Q7aiii The particle comes to rest when $e^{-t} = \frac{1}{4}$,

$\therefore x = 2e^{-2t} + 3e^{-t} + t = 2\left(\frac{1}{4}\right)^2 + 3 \times \frac{1}{4} + \ln 4 = \frac{7}{8} + \ln 4$.

Q12

2010

The acceleration of a particle is given by

$$\ddot{x} = 4 \cos 2t$$

where x is displacement in metres and t is time in seconds.

Initially the particle is at the origin with a velocity of 1 m s^{-1} .

- (i) Show that the velocity of the particle is given by
- (ii) Find the time when the particle first comes to rest.
- (iii) Find the displacement, x , of the particle in terms of t .

$$\dot{x} = 2 \sin 2t + 1.$$

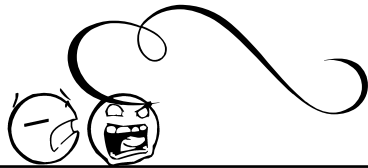
Q7ai $\ddot{x} = 4 \cos 2t$, $\dot{x} = 1$ and $x = 0$ when $t = 0$

$\therefore \dot{x} = \int_0^t 4 \cos 2t dt + 1 = [2 \sin 2t]_0^t + 1 = 2 \sin 2t + 1$

Q7aii Let $\dot{x} = 2 \sin 2t + 1 = 0$,
 $\sin 2t = -0.5$, $2t = \frac{7\pi}{6}$, $\therefore t = \frac{7\pi}{12}$

Q7aiii $\dot{x} = 2 \sin 2t + 1$,

$x = \int_0^t (2 \sin 2t + 1) dt = [-\cos 2t + t]_0^t = 1 - \cos 2t + t$



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Part A: motion



Q13

2011

The velocity of a particle moving along the x -axis is given by

$$\dot{x} = 8 - 8e^{-2t},$$

where t is the time in seconds and x is the displacement in metres.

- Show that the particle is initially at rest.
- Show that the acceleration of the particle is always positive.
- Explain why the particle is moving in the positive direction for all $t > 0$.
- As $t \rightarrow \infty$, the velocity of the particle approaches a constant. Find the value of this constant.
- Sketch the graph of the particle's velocity as a function of time.

Q7bi $\dot{x} = 8 - 8e^{-2t}$

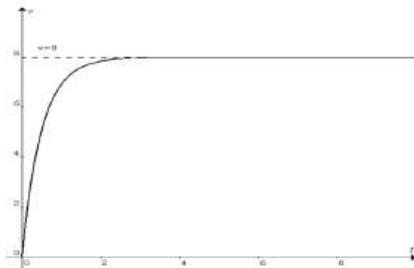
When $t = 0$, $v = \dot{x} = 8 - 8e^0 = 0$, $\therefore$ the particle is initially at rest.

Q7bii $a = \ddot{x} = 16e^{-2t} > 0$, $\therefore$ the acceleration is always positive.

Q7biii The particle starts from rest and its acceleration is always positive, $\therefore$ its velocity is positive for all $t > 0$, i.e. the particle moves in the positive direction for all $t > 0$.

Q7biv $v = \dot{x} = 8 - 8e^{-2t} \rightarrow 8$ as $t \rightarrow \infty$. The value of the constant is 8.

Q7bv



Q14

2012

The velocity of a particle is given by

$$\dot{x} = 1 - 2\cos t,$$

where x is the displacement in metres and t is the time in seconds. Initially the particle is 3 m to the right of the origin.

- Find the initial velocity of the particle.
- Find the maximum velocity of the particle.
- Find the displacement, x , of the particle in terms of t .
- Find the position of the particle when it is at rest for the first time.

Q15bi Assume that 'displacement' means displacement from the origin or simply position (relative to the origin).

$$\dot{x} = 1 - 2\cos t, \text{ at } t = 0, \dot{x} = 1 - 2\cos 0 = -1$$

The initial velocity is 1 m s^{-1} to the left towards the origin.

Q15bii The velocity is maximum when $\cos t = -1$, $\dot{x} = 3 \text{ m s}^{-1}$.

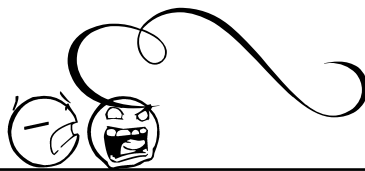
$$\text{Q15biii } x = \int (1 - 2\cos t) dt = t - 2\sin t + c$$

At $t = 0$, $x = 3$, $\therefore c = 3$.

$$\therefore x = t - 2\sin t + 3$$

Q15biv When $\dot{x} = 1 - 2\cos t = 0$, $\cos t = \frac{1}{2}$, $t = \frac{\pi}{3}$.

$$x = \frac{\pi}{3} - 2\sin \frac{\pi}{3} + 3 = \frac{\pi}{3} - \sqrt{3} + 3 \approx 2.32$$



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Part A: motion



Q15

A particle is moving along the x -axis. The displacement of the particle at time t seconds is x metres.

2013

At a certain time, $\dot{x} = -3 \text{ m s}^{-1}$ and $\ddot{x} = 2 \text{ m s}^{-2}$.

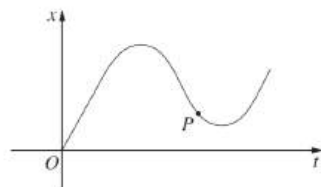
Which statement describes the motion of the particle at that time?

- (A) The particle is moving to the right with increasing speed.
- (B) The particle is moving to the left with increasing speed.
- (C) The particle is moving to the right with decreasing speed.
- (D) The particle is moving to the left with decreasing speed.

Q16

The graph shows the displacement x of a particle moving along a straight line as a function of time t .

2014



Which statement describes the motion of the particle at the point P ?

- (A) The velocity is negative and the acceleration is positive.
- (B) The velocity is negative and the acceleration is negative.
- (C) The velocity is positive and the acceleration is positive.
- (D) The velocity is positive and the acceleration is negative.

Q17

The displacement of a particle moving along the x -axis is given by

2014

$$x = t - \frac{1}{1+t},$$

where x is the displacement from the origin in metres, t is the time in seconds, and $t \geq 0$.

- (i) Show that the acceleration of the particle is always negative.
- (ii) What value does the velocity approach as t increases indefinitely?

Question 13 (c) (i)

$$\begin{aligned} x &= t - \frac{1}{1+t} \\ \dot{x} &= 1 + \frac{1}{(1+t)^2} \\ \ddot{x} &= -\frac{2}{(1+t)^3} \end{aligned}$$

For $t > 0$, $(1+t)^3 > 0$

$$\therefore \ddot{x} = -\frac{2}{(1+t)^3} < 0 \text{ for } t > 0$$

Question 13 (c) (ii)

As $t \rightarrow \infty$, $\dot{x} \rightarrow 1$.

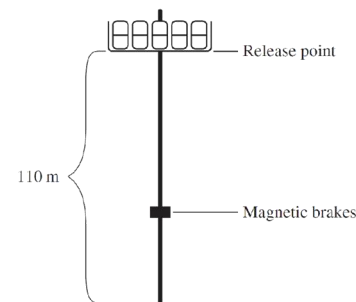
Q18

In a theme park ride, a chair is released from a height of 110 metres and falls vertically. Magnetic brakes are applied when the velocity of the chair reaches -37 metres per second.

2015

The height of the chair at time t seconds is x metres. The acceleration of the chair is given by $\ddot{x} = -10$. At the release point, $t = 0$, $x = 110$ and $\dot{x} = 0$.

- (i) Using calculus, show that $x = -5t^2 + 110$. 2
- (ii) How far has the chair fallen when the magnetic brakes are applied? 2



Question 14 (a) (i)

$$\begin{aligned} \ddot{x} &= -10 \\ \dot{x} &= -10t + c_1 \\ \text{At } t = 0, \dot{x} = 0 &\Rightarrow c_1 = 0 \\ \Rightarrow \dot{x} &= -10t \\ x &= -5t^2 + c_2 \\ \text{At } t = 0, x = 110 &\Rightarrow c_2 = 110 \\ \therefore x &= -5t^2 + 110 \end{aligned}$$

Question 14 (a) (ii)

$$\begin{aligned} \text{At } \dot{x} = -37, -37 &= -10t \Rightarrow t = 3.7 \\ \therefore \text{At } t = 3.7, x &= 41.55 \\ \text{Distance fallen is } &68.45 \end{aligned}$$

Q19

A particle moves in a straight line. Its velocity $v \text{ m s}^{-1}$ at time t seconds is given by $v = 2 - \frac{4}{t+1}$.

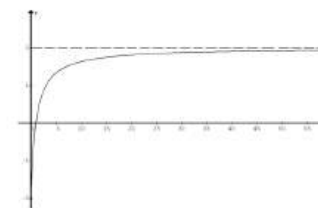
2016

- (i) Find the initial velocity. 1
- (ii) Find the acceleration of the particle when the particle is stationary. 2
- (iii) By considering the behaviour of v for large t , sketch a graph of v against t for $t \geq 0$, showing any intercepts. 2
- (iv) Find the exact distance travelled by the particle in the first 7 seconds. 3

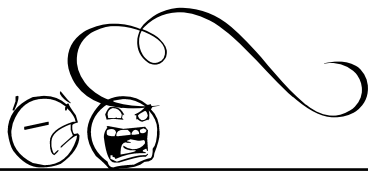
Q16ai $v = 2 - \frac{4}{t+1}$, when $t = 0$, $v = -2$

Q16aai $a = \frac{dv}{dt} = \frac{4}{(t+1)^2}$. When $v = 0$, $t = 1$, $a = 1$.

Q16aiii As $t \rightarrow \infty$, $v \rightarrow 2$

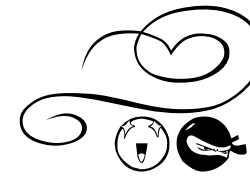


$$\begin{aligned} \text{Q16aiv Distance} &= \left| \int_0^7 \left(2 - \frac{4}{t+1} \right) dt \right| + \left| \int_7^\infty \left(2 - \frac{4}{t+1} \right) dt \right| \\ &= -[2t - 4 \ln(t+1)]_0^7 + [2t - 4 \ln(t+1)]_7^\infty \\ &= -2 + 4 \ln 2 + 14 - 4 \ln 8 - 2 + 4 \ln 2 \\ &= 10 - 4 \ln 2 \text{ metres} \end{aligned}$$



motion and Rates BOS

Part A: motion

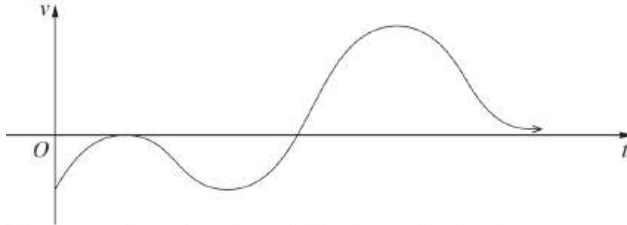


NAME: _____

Q20

2017

A particle is moving along a straight line.
The graph shows the velocity, v , of the particle for time $t \geq 0$.



How many times does the particle change direction?

- A. 1 B. 2 C. 3 D. 4

Q21

2017

Two particles move along the x -axis.

When $t = 0$, particle P_1 is at the origin and moving with velocity 3.

For $t \geq 0$, particle P_1 has acceleration given by $a_1 = 6t + e^{-t}$.

(i) Show that the velocity of particle P_1 is given by $v_1 = 3t^2 + 4 - e^{-t}$.

2

When $t = 0$, particle P_2 is also at the origin.

For $t \geq 0$, particle P_2 has velocity given by $v_2 = 6t + 1 - e^{-t}$.

(ii) When do the two particles have the same velocity?

2

(iii) Show that the two particles do not meet for $t > 0$.

3

Question 15 (c) (i)

Sample answer:

Particle 1 when $t = 0$ $x_1 = 0$ $v_1 = 3$ and $a_1 = 6t + e^{-t}$

v_1 is a primitive of a_1

$$v_1 = \frac{6t^2}{2} + -e^{-t} + k$$

$$= 3t^2 - e^{-t} + k$$

When $t = 0$ $v_1 = 3$

$$3 = 3 \times 0 - e^0 + k$$

$$= -1 + k$$

$$4 = k$$

So $v_1 = 3t^2 + 4 - e^{-t}$

Question 15 (c) (ii)

Sample answer:

$$v_1 = v_2$$

$$3t^2 + 4 - e^{-t} = 6t + 1 - e^{-t}$$

$$3t^2 - 6t + 3 = 0$$

$$3(t^2 - 2t + 1) = 0$$

$$3(t-1)^2 = 0$$

$$t = 1$$

(iii) Show that the particles do not meet for $t > 0$

$$v_1 = 3t^2 + 4 - e^{-t} \text{ so}$$

$$x_1 = \frac{3t^3}{3} + 4t + e^{-t} + C$$

When $t = 0$ $x_1 = 0$

$$0 = 0 + 0 + 1 + C$$

So $C = -1$

$$x_1 = t^3 + 4t + e^{-t} - 1$$

$$v_2 = 6t + 1 - e^{-t}$$

$$x_2 = \frac{6t^2}{2} + t + e^{-t} + k$$

$$= 3t^2 + t + e^{-t} + k$$

When $t = 0$ $x_2 = 0$

$$3 \times 0^2 + 0 + 1 + k = 0$$

$$k = -1$$

So $x_2 = 3t^2 + t + e^{-t} - 1$

If particles meet $x_1 = x_2$

$$\text{So } t^3 + 4t + e^{-t} - 1 = 3t^2 + t + e^{-t} - 1$$

$$t^3 - 3t^2 + 4t - t = 0$$

$$t^3 - 3t^2 + 3t = 0$$

$$t(t^2 - 3t + 3) = 0$$

$t = 0$ (at origin)

$$\text{or } t = \frac{3 \pm \sqrt{9 - 4 \times 3}}{2}$$

$9 - 12 < 0$ so quadratic has no solutions.

$\therefore$ particles do not meet for $t > 0$

(alternative method)

Since both particles start at the origin,

$$v_1 = 3t^2 + 3 + (1 - e^{-t})$$

$$= 3t^2 + 3 + (v_2 - 6t)$$

$$= v_2 + 3(t^2 - 2t + 1)$$

$$= v_2 + 3(t-1)^2$$

So P_1 is never slower than P_2 .

They start together. P_1 starts faster than P_2 and never gets slower.

$\therefore P_1$ will always be ahead of P_2 .

$\therefore$ The particles never meet ($t > 0$).

Q22

2018

The displacement of a particle moving along the x -axis is given by $x = \frac{t^3}{3} - 2t^2 + 3t$,

where x is the displacement from the origin in metres and t is the time in seconds, for $t \geq 0$.

- (i) What is the initial velocity of the particle? 1
- (ii) At which times is the particle stationary? 2
- (iii) Find the position of the particle when the acceleration is zero. 2

Question 12 (d) (i)

Sample answer:

$$x = \frac{t^3}{3} - 2t^2 + 3t$$

$$\dot{x} = t^2 - 4t + 3$$

when $t = 0$

$$\dot{x} = 3 \text{ m/s}$$

Question 12 (d) (ii)

Sample answer:

$$\text{Need } \dot{x} = 0$$

$$t^2 - 4t + 3 = 0$$

$$(t-3)(t-1) = 0$$

$$t = 3, t = 1$$

Question 12 (d) (iii)

Sample answer:

$$\ddot{x} = 2t - 4$$

$$2t - 4 = 0$$

$$2t = 4$$

$$t = 2$$

when $t = 2$

$$x = \frac{2^3}{3} - 2(2)^2 + 3(2)$$

$$= \frac{2}{3} \text{ m}$$

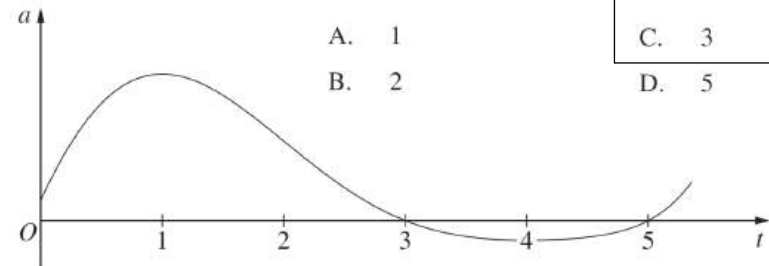
Q23

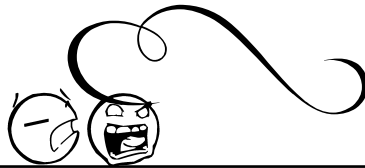
2019

A particle is moving along a straight line. The graph shows the acceleration of the particle.

For what value of t is the velocity v a maximum?

- A. 1 C. 3
- B. 2 D. 5





motion and Rates BOS

Part A: motion



NAME: _____

Q24

A particle is moving along a straight line with displacement x at time t .

2019

The particle is stationary when $t = 11$ and when $t = 13$.

Which of the following **MUST** be true in this case?

- A. The particle changes direction at some time between $t = 11$ and $t = 13$.
- B. The displacement function of the particle has a stationary point at some time between $t = 11$ and $t = 13$.
- C. The acceleration of the particle is 0 at some time between $t = 11$ and $t = 13$.
- D. The acceleration function of the particle has a stationary point at some time between $t = 11$ and $t = 13$.

Q24

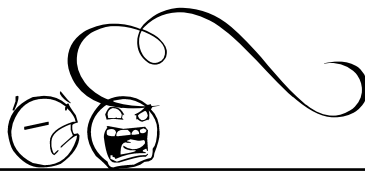
A particle is moving along a straight line. The particle is initially at rest. The acceleration of the particle at time t seconds is given by $a = e^{2t} - 4$, where $t \geq 0$.

2019

Find an expression, in terms of t , for the velocity of the particle.

$$t = 0, v = 0, \frac{dv}{dt} = e^{2t} - 4 \therefore v = \int e^{2t} - 4 dt = \frac{1}{2}e^{2t} - 4t - \frac{1}{2}$$

2



motion and Rates BOS

Part B: Rates



Q 1

2001

When a valve is released, a chemical flows into a large tank that is initially empty. The volume, V litres, of chemical in the tank increases at the rate

$$\frac{dV}{dt} = 2e^t + 2e^{-t}$$

where t is measured in hours from the time the valve is released.

- | | |
|--|---|
| (i) At what rate does the chemical initially enter the tank? | 1 |
| (ii) Use integration to find an expression for V in terms of t . | 2 |
| (iii) Show that $2e^{2t} - 3e^t - 2 = 0$ when $V = 3$. | 1 |
| (iv) Find t , to the nearest minute, when $V = 3$. | 2 |

b) $\frac{dV}{dt} = 2e^t + 2e^{-t}$

(i) At $t = 0$, $\frac{dV}{dt} = 2e^0 + 2e^0$
 $\frac{dV}{dt} = 4$.

$\therefore$ The initial rate is 4 litres per hour.

(ii) $\frac{dV}{dt} = 2e^t + 2e^{-t}$
 $V = 2e^t - 2e^{-t} + c$.

At $t = 0$, $V = 0$.
 $\therefore 0 = 2e^0 - 2e^0 + c$
 $c = 0$
 $\therefore V = 2e^t - 2e^{-t}$.

(iii) When $V = 3$, $3 = 2e^t - 2e^{-t}$.
 Multiply by e^t : $3e^t = 2e^{2t} - 2$.
 That is, $2e^{2t} - 3e^t - 2 = 0$.

(iv) Solve for t , $2(e^t)^2 - 3(e^t) - 2 = 0$.

Let $e^t = x$.

$2x^2 - 3x - 2 = 0$

$(2x+1)(x-2) = 0$

$x = -\frac{1}{2}$ or $x = 2$,

$\therefore e^t = -\frac{1}{2}$ or $e^t = 2$.

$t = \ln\left(-\frac{1}{2}\right)$ or $t = \ln 2$

(no solution) $t = 0.693$ hours
 $= 41.58$ minutes
 ≈ 42 minutes.

Q 2

2002

A cooler, which is initially full, is drained so that at time t seconds the volume of water V , in litres, is given by

$$V = 25\left(1 - \frac{t}{60}\right)^2 \quad \text{for } 0 \leq t \leq 60.$$

- | | |
|---|---|
| (i) How much water was initially in the cooler? | 1 |
| (ii) After how many seconds was the cooler one-quarter full? | 2 |
| (iii) At what rate was the water draining out when the cooler was one-quarter full? | 2 |

(i) $V = 25\left(1 - \frac{t}{60}\right)^2$ for $0 \leq t \leq 60$

$= -\frac{5}{12}$ litres/second.

(Negative value indicates water is draining out.)

At $t = 0$, $V = 25\left(1 - \frac{0}{60}\right)^2$
 $= 25 \times 1$
 $= 25$ litres.

(ii) $V = 25\left(1 - \frac{t}{60}\right)^2$

When $V = \frac{25}{4}$,

$\frac{25}{4} = 25\left(1 - \frac{t}{60}\right)^2$

$\frac{1}{4} = \left(1 - \frac{t}{60}\right)^2$

$\pm \frac{1}{2} = 1 - \frac{t}{60}$

$\frac{t}{60} = 1 \pm \frac{1}{2}$

$\frac{t}{60} = \frac{1}{2}$ or $\frac{t}{60} = \frac{3}{2}$

$\therefore t = 30$ or $t = 90$

$\therefore t = 30$ seconds $0 \leq t \leq 60$.

(ii) Now $\frac{dV}{dt} = 2 \times 25 \left(1 - \frac{t}{60}\right)^1 \cdot \frac{d}{dt} \left(1 - \frac{t}{60}\right)$

$= 50 \left(1 - \frac{t}{60}\right) \cdot -\frac{1}{60}$

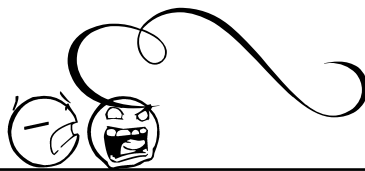
When $t = 30$,

$\frac{dV}{dt} = 50 \left(1 - \frac{30}{60}\right) \times -\frac{1}{60}$

$= 50 \left(\frac{1}{2}\right) \left(-\frac{1}{60}\right)$

$= -\frac{50}{120}$





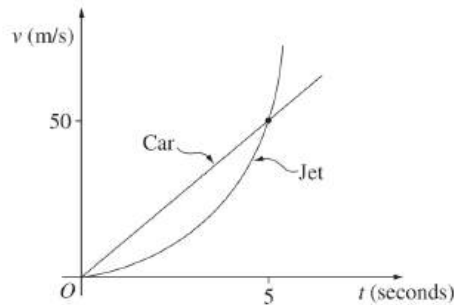
Motion and Rates BOS

Part B: Rates



Q 3

2002



NOT TO SCALE

A car and a jet race one another from rest down a runway. The car increases its speed v_1 at a constant rate, while the speed of the jet is given by $v_2 = 2t^2$. After 5 seconds the car and the jet have the same speed of 50 m/s, as shown on the graph.

- Find an equation for the speed v_1 of the car in terms of t . 1
- How far behind the car is the jet after 5 seconds? 2
- After how many seconds does the jet catch up with the car? 1

(i) The equation of the line for the car is given by

$v_1 = mt$ where m is the slope of the line.

$$\text{Now } m = \frac{50}{5} = 10$$

$$\therefore v_1 = 10t.$$

(ii) Car:

$$v_1 = 10t$$

$$\text{i.e. } \frac{dx_1}{dt} = 10t$$

$$\therefore x_1 = 5t^2 + C$$

$$\text{At } t = 0, x_1 = 0 \therefore C = 0$$

$$\therefore x_1 = 5t^2$$

$$\text{At } t = 5, x_1 = 5 \times 5^2 = 125 \text{ m}$$

Jet:

$$v_2 = 2t^2$$

$$\text{i.e. } \frac{dx_2}{dt} = 2t^2$$

$$\therefore x_2 = \frac{2t^3}{3} + C_1$$

$$\text{At } t = 0, x_2 = 0 \therefore C_1 = 0$$

$$\therefore x_2 = \frac{2t^3}{3}$$

$$\text{At } t = 5, x_2 = \frac{2(5)^3}{3}$$

$$= \frac{250}{3}$$

$$= 83\frac{1}{3} \text{ m.}$$

$$\text{At } t = 5, x_1 - x_2 = 125 - 83\frac{1}{3}$$

$$= 41\frac{2}{3} \text{ m}$$

The jet is $41\frac{2}{3}$ m behind the car.

(iii) The jet catches up with the car when $x_1 = x_2$

$$\therefore 5t^2 = \frac{2}{3}t^3$$

$$5t^2 - \frac{2}{3}t^3 = 0$$

$$t^2\left(5 - \frac{2}{3}t\right) = 0$$

$$\therefore t = 0 \text{ or } 5 - \frac{2}{3}t = 0$$

$$-\frac{2}{3}t = -5$$

$$t = -5 \times -\frac{3}{2}$$

$$= \frac{15}{2}$$

$\therefore$ The jet catches up with car after $7\frac{1}{2}$ s.

Q 4

2005

A tank initially holds 3600 litres of water. The water drains from the bottom of the tank. The tank takes 60 minutes to empty.

A mathematical model predicts that the volume, V litres, of water that will remain in the tank after t minutes is given by

$$V = 3600\left(1 - \frac{t}{60}\right)^2, \text{ where } 0 \leq t \leq 60.$$

- What volume does the model predict will remain after ten minutes? 1
- At what rate does the model predict that the water will drain from the tank after twenty minutes? 2
- At what time does the model predict that the water will drain from the tank at its fastest rate? 2

$$V = 3600\left(1 - \frac{t}{60}\right)^2, \quad 0 \leq t \leq 60.$$

$$\text{(i) When } t = 10, \quad V = 3600\left(1 - \frac{10}{60}\right)^2 = 2500 \text{ L.}$$

$\therefore$ After 10 minutes, 2500 L will remain in the tank.

$$\text{(ii) Rate of draining} = \frac{dV}{dt}$$

$$\frac{dV}{dt} = 3600 \times 2\left(1 - \frac{t}{60}\right) \times \left(-\frac{1}{60}\right) = -120\left(1 - \frac{t}{60}\right).$$

$$\text{When } t = 20, \quad \frac{dV}{dt} = -120\left(1 - \frac{20}{60}\right) = -80.$$

$\therefore$ After 20 minutes, water will drain out of the tank at a rate of 80 L/min.

(iii) Fastest rate when $\frac{dV}{dt}$ is a maximum.

METHOD 1

$$\text{Now } \frac{dV}{dt} = -120\left(1 - \frac{t}{60}\right).$$

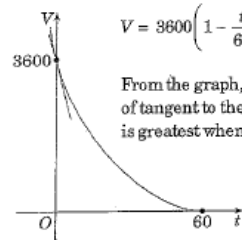
(Note: $0 \leq t \leq 60$)

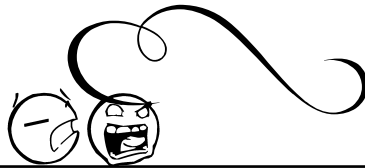
$\therefore$ Maximum draining rate when $t = 0$ (ie. 120 L/min).

METHOD 2

$$V = 3600\left(1 - \frac{t}{60}\right)^2$$

From the graph, slope of tangent to the curve is greatest when $t = 0$. $\left(\frac{dV}{dt}\right)$





motion and Rates BOS

Part B: Rates



Q 5

2006

During a storm, water flows into a 7000-litre tank at a rate of $\frac{dV}{dt}$ litres per minute, where $\frac{dV}{dt} = 120 + 26t - t^2$ and t is the time in minutes since the storm began.

- At what times is the tank filling at twice the initial rate? 2
- Find the volume of water that has flowed into the tank since the start of the storm as a function of t . 1
- Initially, the tank contains 1500 litres of water. When the storm finishes, 30 minutes after it began, the tank is overflowing. 2

How many litres of water have been lost?

(b) (i)

$$\frac{dv}{dt} = 120 + 26t - t^2$$

$$\text{let } t = 0 \therefore \frac{dv}{dt} = 120$$

$$\therefore 240 = 120 + 26t - t^2$$

$$t^2 - 26t + 120 = 0$$

$$(t - 20)(t - 6) = 0$$

$$t = 6 \text{ mins and } 20 \text{ mins}$$

$$\text{when } t = 30$$

$$\therefore v = 120 \times 30 + 13 \times 30^2 - \frac{30^3}{3} + 1500$$

$$v = 7800$$

since the capacity of the tank is

7000 litres there was 800 litres overflowing

(ii)

$$\int \frac{dv}{dt} dt = \int (120 + 26t - t^2) dt$$

$$v = 120t + 13t^2 - \frac{t^3}{3} + c$$

(iii)

$$v = 120t + 13t^2 - \frac{t^3}{3} + c$$

$$\text{when } t = 0 \therefore v = 1500$$

$$c = 1500$$

$$v = 120t + 13t^2 - \frac{t^3}{3} + 1500$$

Q 6

2011

A tap releases liquid A into a tank at the rate of $\left(2 + \frac{t^2}{t+1}\right)$ litres per minute,

where t is time in minutes. A second tap releases liquid B into the same tank at

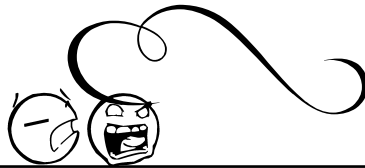
the rate of $\left(1 + \frac{1}{t+1}\right)$ litres per minute. The taps are opened at the same time and

release the liquids into an empty tank.

- Show that the rate of flow of liquid A is greater than the rate of flow of liquid B by t litres per minute. 1
- The taps are closed after 4 minutes. By how many litres is the volume of liquid A greater than the volume of liquid B in the tank when the taps are closed? 2

$$\text{bi } \left(2 + \frac{t^2}{t+1}\right) - \left(1 + \frac{1}{t+1}\right) = 1 + \frac{t^2 - 1}{t+1} = \frac{t^2 + t}{t+1} = \frac{t(t+1)}{t+1} = t$$

$$\text{bii } \Delta V = \int_0^4 t dt = \left[\frac{t^2}{2}\right]_0^4 = 8 \text{ litres}$$



motion and Rates BOS

Part B: Rates



NAME: _____

Q 7

2015

Water is flowing in and out of a rock pool. The volume of water in the pool at time t hours is V litres. The rate of change of the volume is given by

$$\frac{dV}{dt} = 80 \sin(0.5t).$$

At time $t = 0$, the volume of water in the pool is 1200 litres and is increasing.

- | | |
|---|---|
| (i) After what time does the volume of water first start to decrease? | 2 |
| (ii) Find the volume of water in the pool when $t = 3$. | 2 |
| (iii) What is the greatest volume of water in the pool? | 1 |

Question 15 (c) (i)

Consider $\frac{dV}{dt} = 0$
 $80 \sin(0.5t) = 0$
 $\Rightarrow 0.5t = \pi$ (first solution)
 $t = 2\pi$ is where the volume first starts to decrease.

Question 15 (c) (iii)

The volume of water is greatest when $\cos(0.5t) = -1$
 Thus, the greatest volume of water is $-160 + 1360 = 1520$.

Question 15 (c) (ii)

$\frac{dV}{dt} = 80 \sin(0.5t)$
 $\Rightarrow V = -160 \cos(0.5t) + C$
 When $t = 0$, $V = 1200 \Rightarrow C = 1360$
 $\therefore$ At $t = 3$, $V = -160 \cos \frac{3}{2} + 1360 = 1348.68$ (2dp)

Q 8

2017

The rate at which water flows into a tank is given by $\frac{dV}{dt} = \frac{2t}{1+t^2}$, 3

where V is the volume of water in the tank in litres and t is the time in seconds.

Initially the tank is empty. Find the exact amount of water in the tank after 10 seconds.

$$\frac{dV}{dt} = \frac{2t}{1+t^2} \quad \text{when } t=0, V=0$$

$$V = \int_0^{10} \frac{2t}{1+t^2} dt$$

$$= [\ln(1+t^2)]_0^{10}$$

$$= \ln(101) - \ln 1$$

$$= \ln(101) \quad (\approx 4.615...)$$

After 10 seconds the volume of water in the tank is $\ln 101$ litres.

Q 9

2019

The number of leaves, $L(t)$, on a tree t days after the start of autumn can be modelled by

$$L(t) = 200\,000 e^{-0.14t}.$$

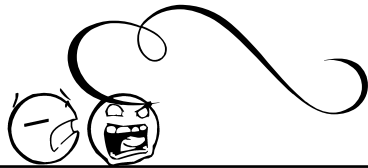
- | | |
|---|---|
| (i) What is the number of leaves on the tree when $t = 31$? | 1 |
| (ii) What is the rate of change of the number of leaves on the tree when $t = 31$? | 2 |
| (iii) For what value of t are there 100 leaves on the tree? | 2 |

ci $L(31) = 200\,000 e^{-0.14 \times 31} \approx 2607$

cii $L'(31) = -0.14 \times 200\,000 e^{-0.14 \times 31} \approx -365$ i.e. dropping 365 leaves per day when $t = 31$

ciii $200\,000 e^{-0.14t} = 100, -0.14t = \log_e \left(\frac{100}{200\,000} \right), t \approx 54.3$





motion and Rates BOS

Part B: Rates



NAME: _____

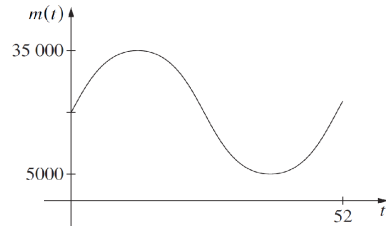
Q10

2020

The population of mice on an isolated island can be modelled by the function

$$m(t) = a \sin\left(\frac{\pi}{26}t\right) + b,$$

where t is the time in weeks and $0 \leq t \leq 52$. The population of mice reaches a maximum of 35 000 when $t = 13$ and a minimum of 5000 when $t = 39$. The graph of $m(t)$ is shown.



- (a) What are the values of a and b ? 2
- (b) On the same island, the population of cats can be modelled by the function 3

$$c(t) = -80 \cos\left(\frac{\pi}{26}(t - 10)\right) + 120.$$

Consider the graph of $m(t)$ and the graph of $c(t)$.

Find the values of t , $0 \leq t \leq 52$, for which both populations are increasing.

- (c) Find the rate of change of the mice population when the cat population reaches a maximum. 2

(a) $a = \frac{35\,000 - 5000}{2}$
 $= 15\,000$
 $b = \frac{35\,000 + 5000}{2}$
 $= 20\,000$

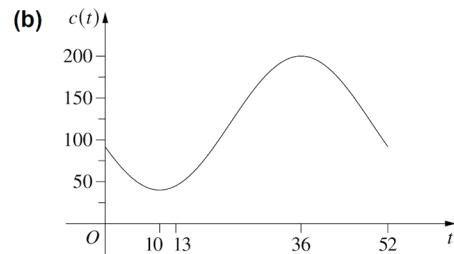
- (c) The cat population reaches a maximum at $t = 36$.

$$m'(t) = \frac{\pi}{26} \times 15\,000 \cos\left(\frac{\pi}{26}t\right)$$

$$m'(36) = \frac{\pi}{26} \times 15\,000 \cos\left(\frac{\pi}{26} \times 36\right)$$

$$= -642.7\dots$$

Therefore the mice population is decreasing at 643 mice per week.



The population of mice increases for $0 < t < 13$ and $39 < t < 52$.

The population of cats increases for $10 < t < 36$.

Therefore both populations are increasing for $10 < t < 13$.

